Zhifei Li

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Research Interests

My research interests lie in **designing efficient systems for ML**, focusing on cloud resource orchestration, distributed training infrastructure, and compound AI systems addressing the growing resource demands of diverse AI applications. I am also interested exploring in how AI techniques can advance systems design.

Research Experience

Sky Computing Lab — University of California, Berkeley

Research Intern, advised by Prof. Ion Stoica
worked with Prof. Joseph E. Gonzalez, Prof. Matei Zaharia — July 2025 - December 2025

- **SkyNomad: Multi-Region Spot Instance Scheduling** (accepted to **NSDI '27**)
  - Designed a multi-region spot instance scheduling system, addressing single-region availability bottlenecks for offline workloads, via Unified Cost Model trading off cross-region availability and pricing vs. migration costs
  - Achieved >50% cost reduction over previous SOTA (NSDI '24 best paper), saved $1,000+ from a $2,200 training job vs. AWS SageMaker
  - Led project from research formulation to production, drove methodology design, and built simulation framework
- **LEANN: Storage-Efficient Compound AI Systems** (**MLSys '26**)
  - Co-designed a two-level recompute algorithm to cut vector index storage overhead in RAG pipelines
  - Achieved 97% storage reduction, <5% latency impact; led open-source implementation to 9,000+ GitHub stars
  - Built the system extending `FAISS` C++, contributing 70% codebase; drove evaluation efforts
- **AI-driven Systems Research**
  - Investigated systems optimization through evolutionary algorithms and LLM-guided design space exploration
  - Led case study in Barbarians at the Gate paper, which demonstrated 30% improvement over the SOTA
  - Co-developed FrontierCS benchmark with problem specifications and evaluations for 40 open-ended problems

Education

University of California, Berkeley — Berkeley, CA, USA

Exchange Student, Computer Science — August 2024 - December 2024

- **CS294-162 Machine Learning Systems graduate seminar**
  - Optimized complex DAG workload execution through intelligent data placement and cross-cloud task scheduling
  - Achieved 45% cost reduction; select optimizations merged into SkyPilot

Renmin University of China (Ranked 23rd globally on CSRankings 2025) — Beijing, China

Bachelor's in Computer Science, **Turing Honors Class** — September 2022 - Expected: 06/26

- GPA: 3.8/4.0 (Top 5%)

Publications

SkyNomad: On Using Multi-Region Spot Instances to Minimize AI Batch Job Cost
Zhifei Li*, *Tian Xia**, *Ziming Mao*, *Zihan Zhou*, *Ethan J. Jackson*, *Jamison Kerney*, *Zhanghao Wu*, *Pratik Mishra*, *Yi Xu*, *Yifan Qiao*, *Ion Stoica*
NSDI '27; [PDF]

LEANN: A Low-Storage Overhead Vector Index
Yichuan Wang, **Zhifei Li**, *Shu Liu*, *Yongji Wu*, *et al.*, *Ion Stoica*, *Sewon Min*, *Matei Zaharia*, *Joseph Gonzalez*
MLSys '26 (Best Paper Award); [PDF]

Barbarians at the Gate: How AI is Upending Systems Research
*Audrey Cheng**, *Shu Liu**, *Melissa Pan**, **Zhifei Li**, *Bowen Wang*, *Alex Krentsel*, *et al.*, *Matei Zaharia*, *Ion Stoica*
arXiv preprint; [PDF]

FrontierCS: Evolving Challenges for Evolving Intelligence
Qiuyang Mang, Wenhao Chai*,* **Zhifei Li***, *Huanzhi Mao*, et al., Ion Stoica, Jingbo Shang, Zhuang Liu, Alvin Cheung*
ICML '26; [PDF]; [Website]

SkyWalker: A Locality-Aware Cross-Region Load Balancer for LLM Inference
Tian Xia, Ziming Mao, Jamison Kerney, Ethan J. Jackson, **Zhifei Li**, *Jiarong Xing, Scott Shenker, Ion Stoica*
EuroSys 2026; [PDF]

Open-Source Projects

LEANN: the Smallest Vector Index in the World — LEANN 9k ⭐
Enjoy 97% storage savings for RAG application on your personal device — September 2024 - Present
- Led research-to-production translation of LEANN from prototype to production-ready open-source Python package with CI/CD pipeline, grew to 9,000+ GitHub stars with 20 active external contributors and 40k+ downloads
- Drove technical outreach including blog posts social media campaign achieving 600k+ views

SkyPilot: Run AI on Any Infra — SkyPilot 9.1k ⭐
Framework for running ML/AI workloads across any cloud infrastructure — September 2024 - Present
- Top 10 contributor; created 70+ issues and merged 50+ pull requests; contributed 30,000+ lines of code changes
- Implemented High Availability Controller for SkyServe control plane; adopted by startups including Hypermode, IP Copilot and Liner

Services

USENIX ATC '25 Artifact Evaluation Committee
Reviewer — May 2025

Introduction to Computer Systems (ICS)
Head Teaching Assistant — Fall 2024, Spring 2025
- Led 6 TAs in teaching systems curriculum that covered cache hierarchies and memory optimization to 200+ students

RUC Computer Association
President — July 2024 - July 2025
- Organized 10+ tech talks, 5000+ total attendees, covering topics from Functional Programming to Rust ecosystem
- Led 100+ members across 6 departments, fostered a startup atmosphere and inclusive environment

Cheese Tech
Co-founder — September 2023 - October 2024
- Raised $300K seed funding, grew to 1,000+ users across 3 partner institutions within first year
- Developed AI-powered research platform with inspiration tracking, progress management, and intelligent advising

Honors and Awards

- ***First Place***, *National System Capability Competition* (among **10,000** from 1,200 schools) — November 2025
- *Elite Collegiate Award, China Computer Federation* (<100 recipients **nationally**) — August 2025
- *Dean's Scholarship, Gaoling School of AI* (15 recipients out of 2,000) — May 2025
- *National Scholarship* (Top 0.2% nationally) — September 2024
- *First-Class Scholarship for Social Service* (48 recipients out of **35,000**) — September 2023
- *First Prize, National Olympiad in Informatics, Beijing* (300 winners) — December 2019

Skills

- **Coding** **C++**, **CUDA**, **Python**, Rust, TypeScript
- **Tools** **PyTorch**, **SkyPilot**, Ray, Kubernetes, verl, NeMo, Nix, Fish, Typst
- **Languages** English, Chinese (native), French
- **Hobbies** Texas Hold'em, Badminton, Chess, Cooking